

## Multiple Choice (10 marks)

1. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
  - (a) \*\*
  - (b) //
  - (c) is
  - (d) not in
2. Which of the following has the greatest potential for compromising a user's personal privacy?
  - (a) A group of cookies stored by the user's Web browser
  - (b) The Internet Protocol (IP) address of the user's computer
  - (c) The user's e-mail address
  - (d) The user's public key used for encryption
3. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation-free?
  - (a) Round-robin
  - (b) Priority queuing
  - (c) Shortest job first
  - (d) Youngest job first
4. Which of the following is the hexadecimal representation of the decimal number  $231_{10}$ ?
  - (a)  $17_{16}$
  - (b)  $E4_{16}$
  - (c)  $E7_{16}$
  - (d)  $F4_{16}$
5. Let  $l = [1,2,3,4]$ . What is  $\min(l)$  in Python 3?
  - (a) 1
  - (b) 2
  - (c) 3
  - (d) 4

## True / False (10 marks)

*Answer True or False*

6. True or False: The set of Boolean operators {AND, OR} is not complete, as it cannot represent all Boolean expressions.
7. True or False: In Python 3, the expression  $x \gg 1$  evaluates to 3 when  $x$  is assigned the value 8.
8. True or False: The maximum possible degree of the minimal-degree interpolating polynomial  $p(x)$  for  $n + 1$  points is  $n + 1$ .
9. True or False: The language  $\{ww \mid w \in (0 + 1)^*\}$  is accepted by some Turing machines, but not by any pushdown automaton.
10. True or False: A heap area is usually not represented in a subroutine's activation record frame in stack-based programming languages.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function for computing square roots using the babylonian method.
12. Write a function to find the longest common prefix in the given set of strings.

*End of Paper*